

**Systems Genetics**  
**Spring 2026**

**Lecture 10**

**Polygenic Risk Scores**  
**(PRS)**

Irit Gat-Viks

# Outline

PRS definitions

PRS – classical definition

PRS – shrinkage

PRS – Bayesian

Presentation of PRS results

The clinical utility of PRS

PRS in systems genetics

# Part 1:

## PRS definition

# Polygenic risk score (PRS)

**Polygenic risk scores**, or polygenic scores, are defined as a single value estimate of an individual's genetic relevance to a phenotype.

## A very naïve definition of PRS

- Use machine learning algorithm to forecast the trait (a continuous outcome,  $y$ ) based on the values of many genetic variants (multiple predictor variables,  $x$ ).
- For instance: multiple regression analysis, SVM, random forest, (deep) neural networks...
- The problem: a huge search space, a high risk of overfitting and lack of generalization.
- Random effect is irrelevant, as we need predictor (known effects for prediction in new individuals)

## The classical definition of PRS

- PRS are calculated as a sum of their genome-wide genotypes, weighted by corresponding genotype effect size estimates derived from GWAS summary statistic data.
- Given the summary statistics from GWAS, there is no need to perform any additional learning from data (no overfitting).
- Method parameters: Use thresholding on summary statistics of GWAS

# Example of classical definition of PRS



April 2020

## Polygenic and clinical risk scores and their impact on age at onset and prediction of cardiometabolic diseases and common cancers

Nina Mars<sup>1</sup>, Jukka T. Koskela<sup>1</sup>, Pietari Ripatti<sup>1</sup>, Tuomo T. J. Kiiskinen<sup>1</sup>, Aki S. Havulinna<sup>1,2</sup>, Joni V. Lindbohm<sup>3</sup>, Ari Ahola-Olli<sup>1</sup>, Mitja Kurki<sup>1,4,5</sup>, Juha Karjalainen<sup>1,6,7</sup>, Priit Palta<sup>1,8</sup>, FinnGen\*, Benjamin M. Neale<sup>4,7</sup>, Mark Daly<sup>1,6</sup>, Veikko Salomaa<sup>2</sup>, Aarno Palotie<sup>1,5,6</sup>, Elisabeth Widén<sup>1</sup> and Samuli Ripatti<sup>1,3,6</sup> ✉

We calculated the PRSs by summing the dosage of each risk allele carried by an individual (ranging from 0 to 2 for each variant, dosage used for incorporating imputation uncertainty), weighting each variant by its natural logarithm of the relative risk extracted from the GWAS. For each individual  $i$ , this results in a single value on a continuous scale:

$$\text{PRS}_i = \sum_{j=1}^M \hat{\beta}_j \times \text{dosage}_{ij}$$

where  $\hat{\beta}_j$  is the weight for variant  $j$  obtained from GWAS summary statistics.

## Relative Risk (RR)

- To calculate the risk associated with an exposure, we must compare the risk (incidence) among the exposed to those not exposed

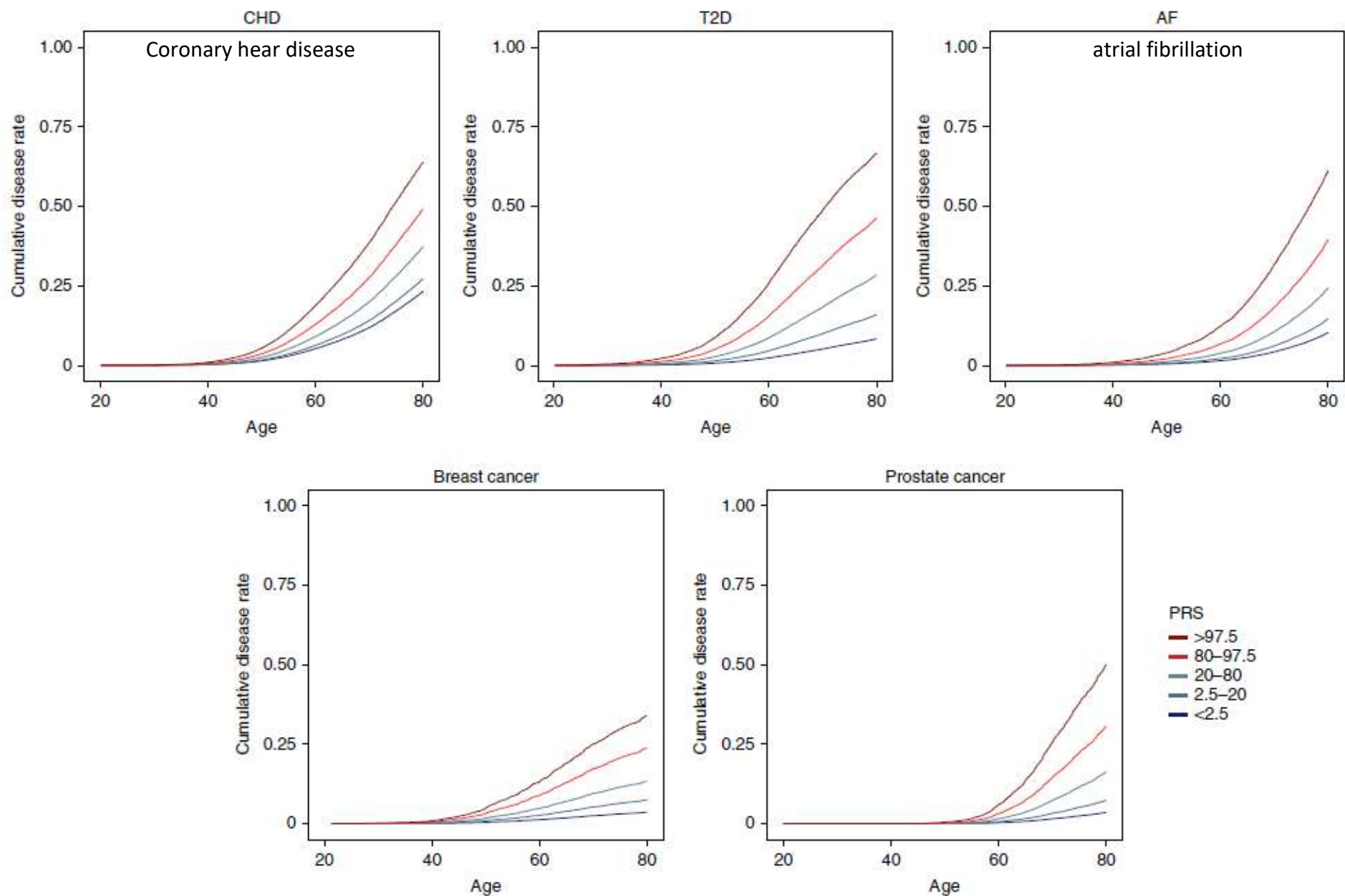
$$\text{Ratio of Risks} = \frac{\text{Disease Risk (incidence) in Exposed (A/A+B)}}{\text{Disease Risk (incidence) in Non-Exposed (C/C+D)}} = \text{Relative Risk}$$

|           | Disease | No Disease |
|-----------|---------|------------|
| Exposed   | A       | B          |
| Unexposed | C       | D          |

## Calculating the Odds Ratio (OR)

|           | Disease (Case) | No Disease (Control) |
|-----------|----------------|----------------------|
| Exposed   | A              | B                    |
| Unexposed | C              | D                    |

$$\text{OR} = \frac{\text{Odds that a case was exposed (A/C)}}{\text{Odds that a control was exposed (B/D)}}$$



**Fig. 1 | Adjusted survival curves from Cox proportional hazards models showing the cumulative risk of disease by PRS category in FinnGen ( $n = 135,300$  individuals).  $P$  values for trend: CHD  $P = 2.80 \times 10^{-256}$ ; T2D  $P < 1.00 \times 10^{-300}$ ; AF  $P < 1.00 \times 10^{-300}$ ; breast cancer  $P = 3.07 \times 10^{-183}$ ; prostate cancer  $P = 2.41 \times 10^{-243}$ . Incident and prevalent cases are included. All tests were two-tailed.**

## **Improved PRS - shrinkage:**

The challenge: Given that SNP effects are estimated with uncertainty, and since not all SNPs influence the trait under study, the use of effect size estimates of all SNPs could generate poorly estimated PRSs with high standard error.

The solution: Shrinkage of the effect size of SNPs based on genotype/phenotype target data



# Regularization to avoid overfitting

מודל הרגרסיה הליניארית המרובה:

מקדמי  $\beta_0, \beta_1, \dots, \beta_{k-1}$  הם  $y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \dots + \beta_k x_{i,k-1} + \varepsilon_i$  כאשר  
הרגרסיה  $\varepsilon_i$  הנם רכיבי שגיאה ב"ת שווי התפלגות -  $N(0, \sigma^2)$ .

פתרון הרגרסיה מביא למינימום את סכום ריבועי השגיאות:

$$\hat{\beta}_{\text{MLE}} = \arg \min_{\beta} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2 = \arg \min_{\beta} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \longrightarrow \text{The "loss" functions}$$

Regularization: modifying the loss function with a penalty term which effectively **shrinks** the estimates of the coefficients. These types of methods within the framework of regression are also called “shrinkage” methods or “penalized regression” methods.

The “loss” functions

$$\hat{\beta}_{\text{L1}} = \arg \min_{\beta} \left( \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2 \right)$$

Promote smaller coefficients; sparsity of coefficients (some are zero); reduce overfitting; can be seen as a form of feature selection.

# Max-likelihood perspective of regression

- Max likelihood estimator == Least squares regression

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \log P(y|\theta)$$

$$\begin{aligned} P(\mathbf{y}|\beta) &= \prod_{i=1}^n P_Y(y_i|\beta, \sigma^2) \\ &= \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2}{2\sigma^2}} \end{aligned}$$

$$\begin{aligned} &\arg \max_{\beta} \log \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2}{2\sigma^2}} \\ &= \arg \max_{\beta} - \sum_{i=1}^n \frac{(y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2}{2\sigma^2} \\ &= \arg \min_{\beta} \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2 \\ &= \arg \min_{\beta} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2 . \end{aligned}$$

The least squares  
cost function

# Bayesian inference

In Bayesian inference, we're primarily concerned with the posterior.

The **posterior**: The probability of the parameters given the data.

$$P(\theta|y) = \frac{P(y|\theta)P(\theta)}{P(y)}$$
$$\text{posterior} = \frac{\text{likelihood} \cdot \text{prior}}{\text{evidence}}$$

The goal is to find the **maximum of the posterior** – i.e., the maximum a posteriori probability estimate (MAP).

$$\begin{aligned}\hat{\theta}_{\text{MAP}} &= \arg \max_{\theta} P(\theta|y) \\ &= \arg \max_{\theta} \frac{P(y|\theta)P(\theta)}{P(y)} \\ &= \arg \max_{\theta} P(y|\theta)P(\theta) \\ &= \arg \max_{\theta} \log(P(y|\theta)P(\theta)) \\ &= \boxed{\arg \max_{\theta} \log P(y|\theta)} + \boxed{\log P(\theta)}\end{aligned}$$

MLE estimator  
(least squares in  
linear regression)

Bayesian priors on the  
coefficients of linear  
regression

# Selecting Priors for Linear Regression

Normally Distributed Priors:  $\beta_j \sim \mathcal{N}(0, \tau)$

MLE estimator

$$\arg \max_{\beta} \left[ \log \prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2}{2\sigma^2}} \right]$$

# Selecting Priors for Linear Regression

**Laplacean-Distributed Priors:**  $\beta_j \sim \text{Laplace}(0, b)$   
(location parameter and scale parameter)

$$\text{Laplace}(\mu, b) = \frac{1}{2b} e^{-\frac{|x-\mu|}{b}}$$

$$\begin{aligned} & \arg \max_{\beta} \left[ \log \prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2}{2\sigma^2}} + \log \prod_{j=0}^p \frac{1}{2b} e^{-\frac{|\beta_j|}{b}} \right] \\ &= \arg \max_{\beta} \left[ -\sum_{i=1}^n \frac{(y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2}{2\sigma^2} - \sum_{j=0}^p \frac{|\beta_j|}{b} \right] \\ &= \arg \min_{\beta} \frac{1}{2\sigma^2} \left[ \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2 + \frac{2\sigma^2}{b} \sum_{j=0}^p |\beta_j| \right] \\ &= \arg \min_{\beta} \left[ \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}))^2 + \lambda \sum_{j=0}^p |\beta_j| \right] \end{aligned}$$

L1 regularization  
(LASSO)

# Example of improved PRS



ARTICLE

<https://doi.org/10.1038/s41467-019-09718-5>

OPEN

## Polygenic prediction via Bayesian regression and continuous shrinkage priors

Tian Ge<sup>1,2,3</sup>, Chia-Yen Chen<sup>1,2,3,4</sup> , Yang Ni<sup>5</sup> , Yen-Chen Anne Feng<sup>1,2,3,4</sup> & Jordan W. Smoller<sup>1,2,3</sup>

This PRS relies on **Bayesian regression framework**, using a prior on SNP effect sizes. It is a marker-specific adaptive shrinkage – i.e., the amount of shrinkage applied to each genetic marker is adaptive to the strength of its association signal in GWAS.

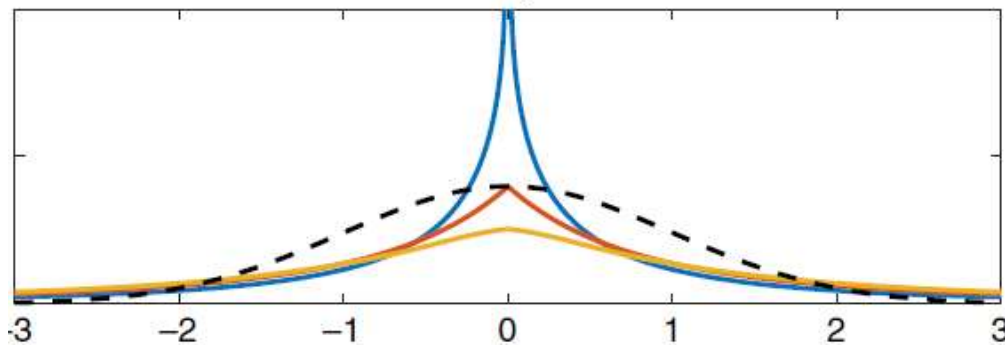


**Conceptual frameworks.** We consider a Bayesian high-dimensional regression framework for polygenic modeling and prediction:

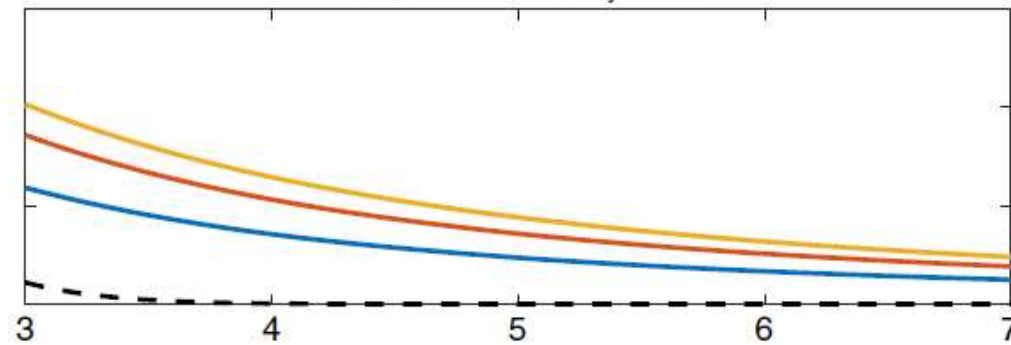
$$\mathbf{y}_{N \times 1} = \mathbf{X}_{N \times M} \boldsymbol{\beta}_{M \times 1} + \boldsymbol{\varepsilon}_{N \times 1}, \quad (1)$$

where  $N$  and  $M$  denote the sample size and number of genetic markers, respectively,  $\mathbf{y}$  is a vector of traits,  $\mathbf{X}$  is the genotype matrix,  $\boldsymbol{\beta}$  is a vector of effect sizes for the genetic markers, and  $\boldsymbol{\varepsilon}$  is a vector of residuals. By assigning appropriate priors on the regression coefficients  $\boldsymbol{\beta}$  to impose regularization, additive PRS can be calculated using posterior mean effect sizes.

Prior density of  $\beta_j$ : central region



Prior density of  $\beta_j$ : tails



- A SNP with weak association in GWAS
- A SNP with mid level of association in GWAS
- A SNP with a strong association in GWAS

\* All priors densities are a mixture of several gaussians with different weights



# The Polygenic Score (PGS) Catalog

An open database of polygenic scores and the relevant metadata required for accurate application and evaluation.



Examples: [breast cancer](#), [glaucoma](#), [BMI](#), [EFO\\_0001645](#)

## Explore the Data

In the current PGS Catalog you can **browse** the scores and metadata through the following categories:

Polygenic Scores

⌚ 2,199

Traits

🧑 538

Publications

📖 318



Part 2:

The clinical utility of PRS

# References

nature  
protocols

REVIEW ARTICLE

<https://doi.org/10.1038/s41596-020-0353-1>

 Check for updates

## Tutorial: a guide to performing polygenic risk score analyses

Shing Wan Choi<sup>1,2</sup>, Timothy Shin-Heng Mak<sup>3</sup>  and Paul F. O'Reilly<sup>1,2</sup> 

 TRANSLATIONAL GENETICS

## The personal and clinical utility of polygenic risk scores

Ali Torkamani<sup>1,2</sup> \*, Nathan E. Wineinger<sup>1,2</sup> and Eric J. Topol<sup>1,3</sup>

## Viewpoint, Nature reviews genetics 2022, Nilanjan Chatterjee

Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, USA.

*“There is widespread misunderstanding around the utility of PGS for **population risk stratification** versus **individualized risk prediction**. PGSs for several common diseases, such as breast cancer, now have the ability to stratify a population into distinct categories of risk for which the risk–benefit balance for existing interventions is sufficiently different. However, these same PGSs have modest utility when it comes to accuracy of individual-level prediction or risk discrimination. Although this may seem contradictory, the fact is that a model need not have very high individual-level prediction accuracy to have a meaningful population-level impact for guiding interventions”.*

### **Despite modest predictive ability, PGS can be useful to identify:**

- Small high-risk populations who would benefit from certain interventions that may not be recommended for the population at large because of risk–benefit balance.
- Small low-risk populations who could avoid or delay interventions that have substantial cost or harm associated with them.

Most cases of a given disease will occur outside small high- risk groups

Stratification/subtypes based on PGS categories can be a useful approach in precision medicine

### Example 1: using a combination of only 2 genetic variants: *BRCA1* and *BRCA2*

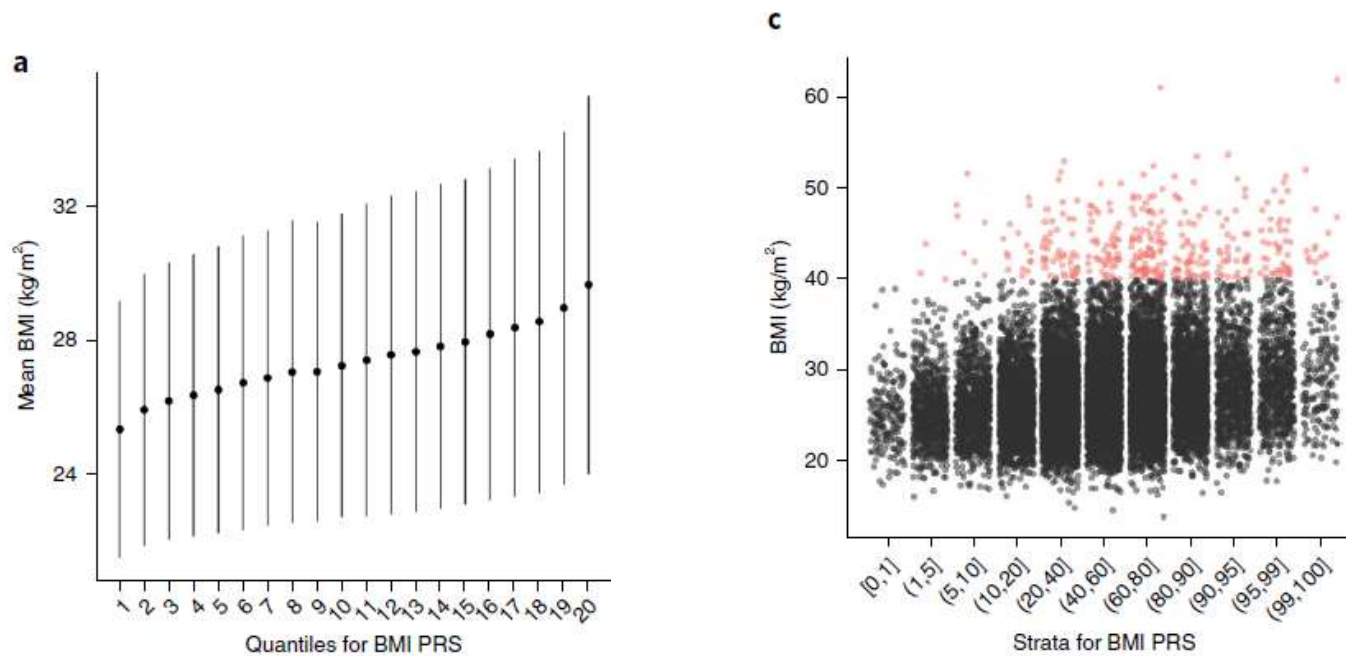
The prevalence of *BRCA1* and *BRCA2* mutations is  $\ll 1\%$  in the population.

**Poor individual-level prediction:** the total incidence of breast cancer attributable to *BRCA1* and *BRCA2* mutations are low, accounting for approximately 5% of all breast cancer cases.

**Utility in population risk stratification:** Pathogenic *BRCA1* and *BRCA2* mutations confer an estimated 65% and 45% absolute lifetime risk of developing breast cancer, respectively, in contrast to the 12% absolute lifetime risk in the general population.  
→ *BRCA1* and *BRCA2* testing can identify a subset of individuals whose absolute risk of disease is significantly higher than that of the average individual in the general population. These high-risk individuals could benefit from a tailored health management strategy, which may include intensive screening or more invasive interventions.

The NCI BCRAT model for breast cancer (also known as Gail model), includes only a handful of risk factors and has modest discriminatory utility (area under the curve (AUC)  $< 60\%$ ), has been shown to have the ability to stratify a population into risk categories for which the risk–benefit balance of chemoprevention can differ substantially.

## Example 2: PRSs calculated in 386,266 individuals in the UK Biobank, derived using the GIANT BMI GWAS



**Poor individual-level prediction:** PRS for BMI explains only 5% of the variation in BMI. The uncertainty in the individual predictions is extremely large, such that individuals should avoid interpretation of their PRS value.

**Utility in population risk stratification:** the upper PRS strata do have an elevated prevalence of obesity with implications on guiding of interventions.

**Caution:** Most obese individuals have a normal or low PRS, highlighting a drawback of focusing on 'high risk' strata. Such 'high risk' strata are misleading in terms of impact on the public health.

# Clinical utility - example

- Estimation of the residual lifetime risk for diseases for which risk-adjusted screening is meaningful: breast cancer
- The American Cancer Society recommends MRI screening when the lifetime risk exceeds 20–25%.
- CanRisk (Carver et al. 2021) calculates BOADICEA version 6 which includes the PRS of Mavaddat et al. comprising 313 SNPs (in practical use in Germany).

Journal of Community Genetics  
<https://doi.org/10.1007/s12687-023-00645-z>

RESEARCH

**Clinical utility of polygenic risk scores: a critical 2023 appraisal**

Sebastian Koch<sup>1</sup> · Jörg Schmidtke<sup>2,3</sup> · Michael Krawczak<sup>1</sup> · Amke Caliebe<sup>1</sup>

[nature](#) > [nature genetics](#) > [articles](#) > [article](#)

Article | [Open access](#) | Published: 30 April 2026

## Genetic association and machine learning improve the prediction of type 1 diabetes risk

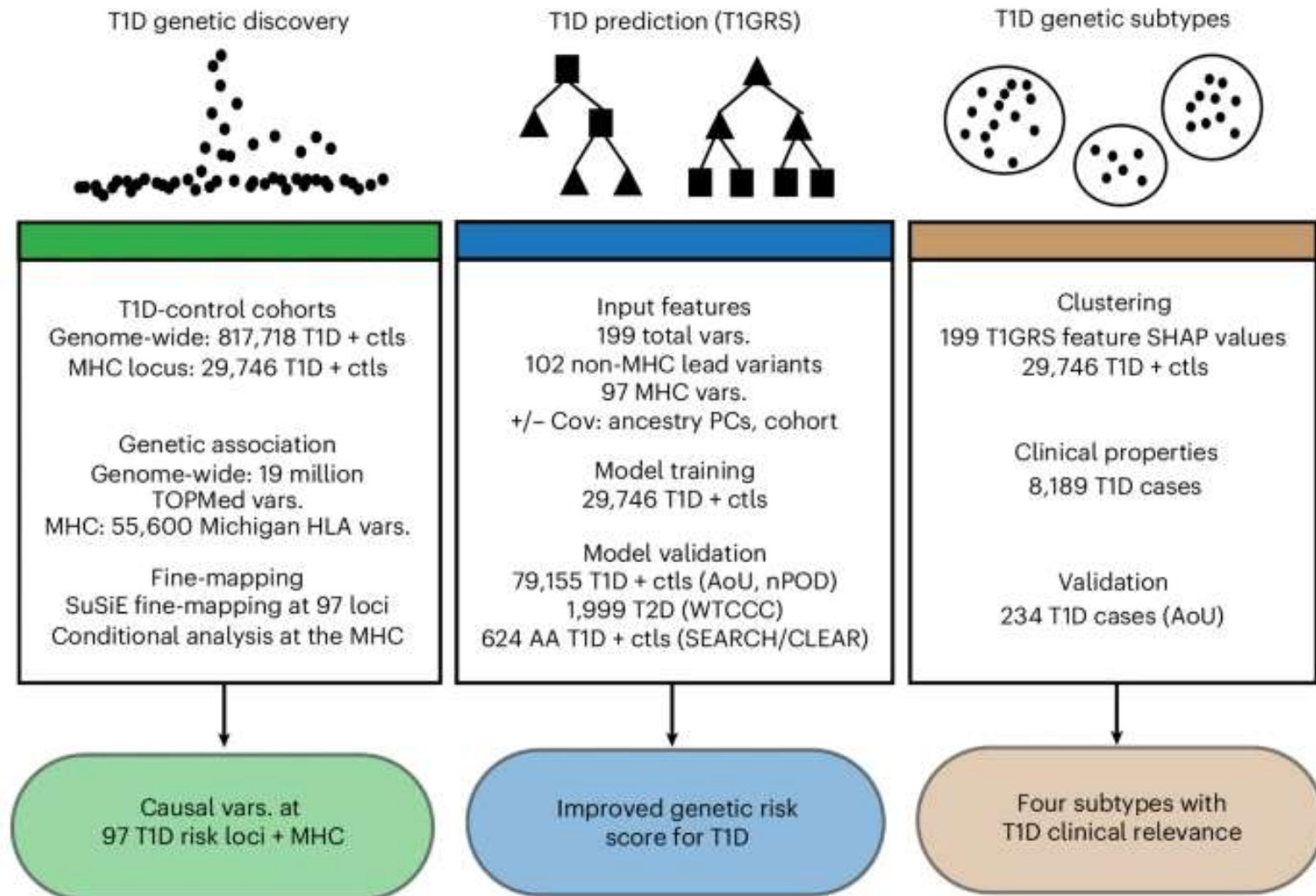
[Carolyn McGrail](#) , [Timothy J. Sears](#), [Emily N. Griffin](#), [Alexandra L. Ghaben](#), [Patrick Smadbeck](#), [Jason Flannick](#), [Parul Kudtarkar](#), [Hannah Carter](#) & [Kyle Gaulton](#) 

*Nature Genetics* **58**, 1062–1072 (2026) | [Cite this article](#)

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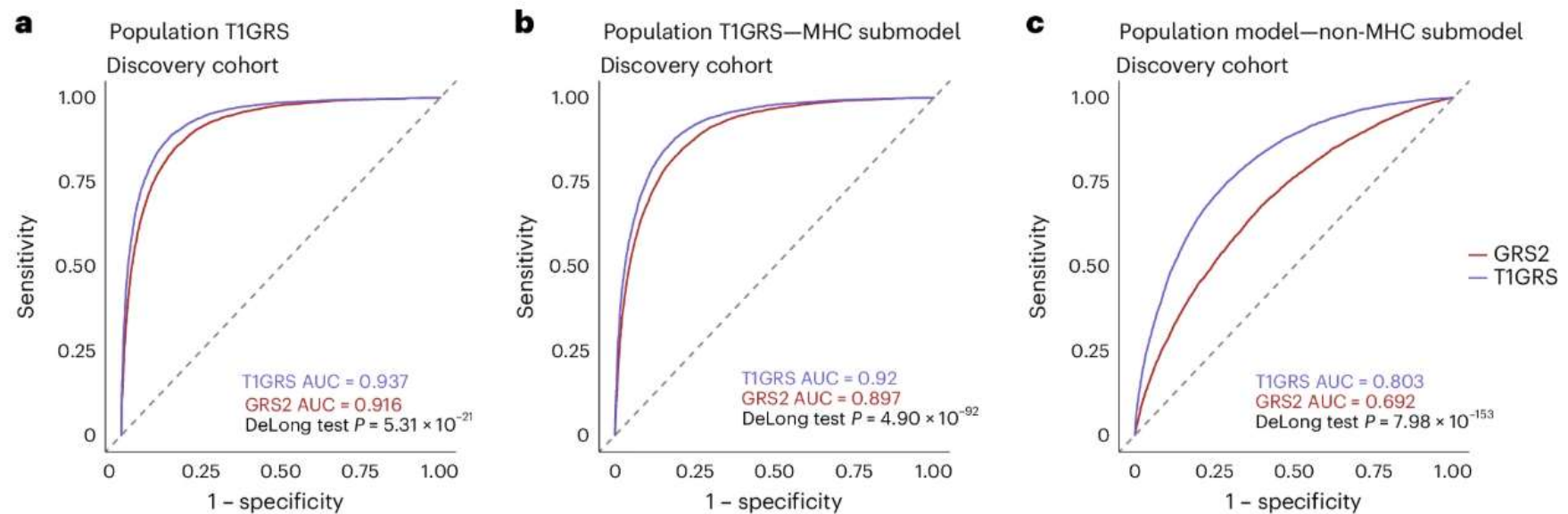


**Fig. 1: Overview of study design.**



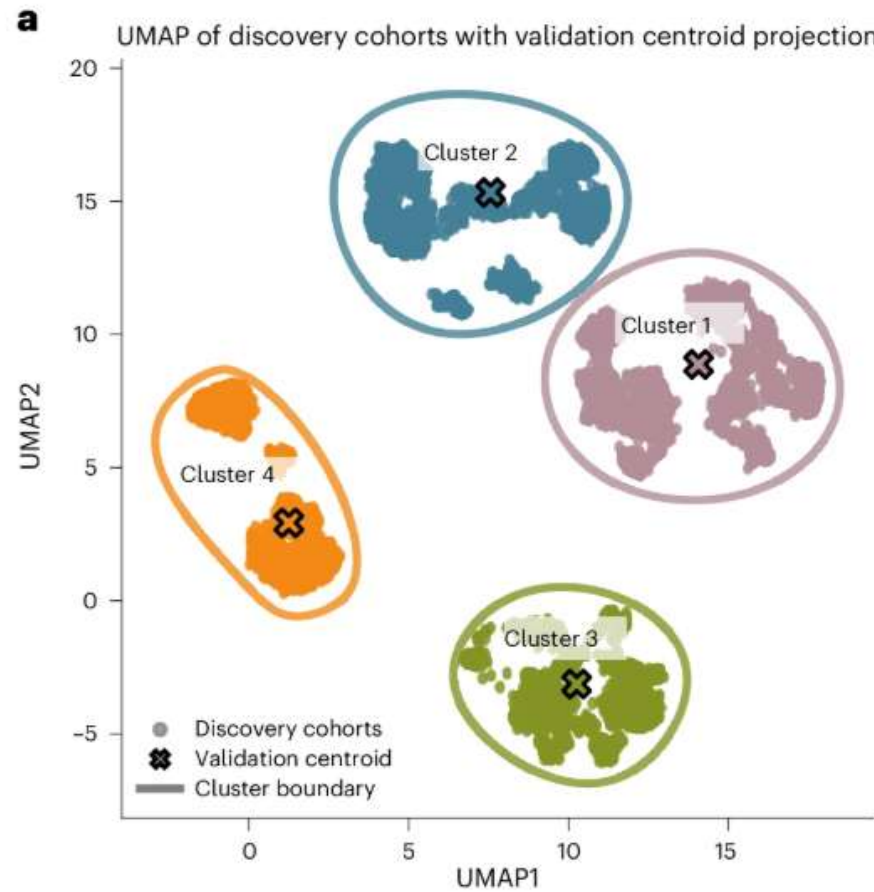
## Fig. 4: Prediction of T1D using machine learning of variant genotypes.

From: [Genetic association and machine learning improve the prediction of type 1 diabetes risk](#)



## Fig. 6: T1GRS reveals clinically relevant genetic subtypes of T1D.

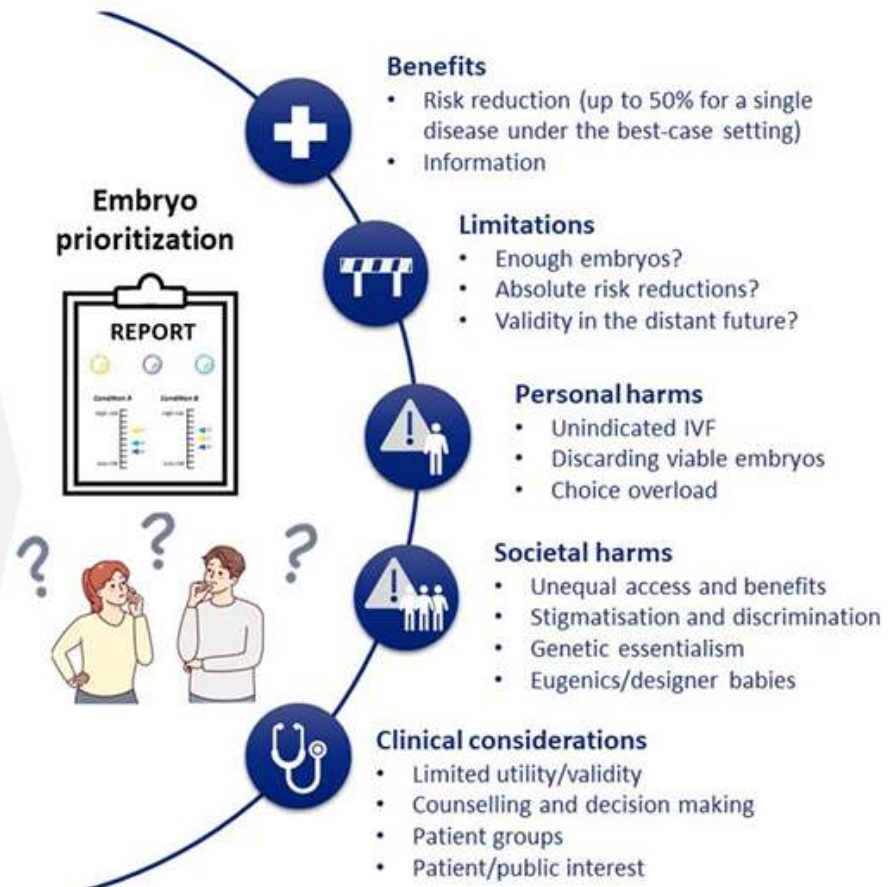
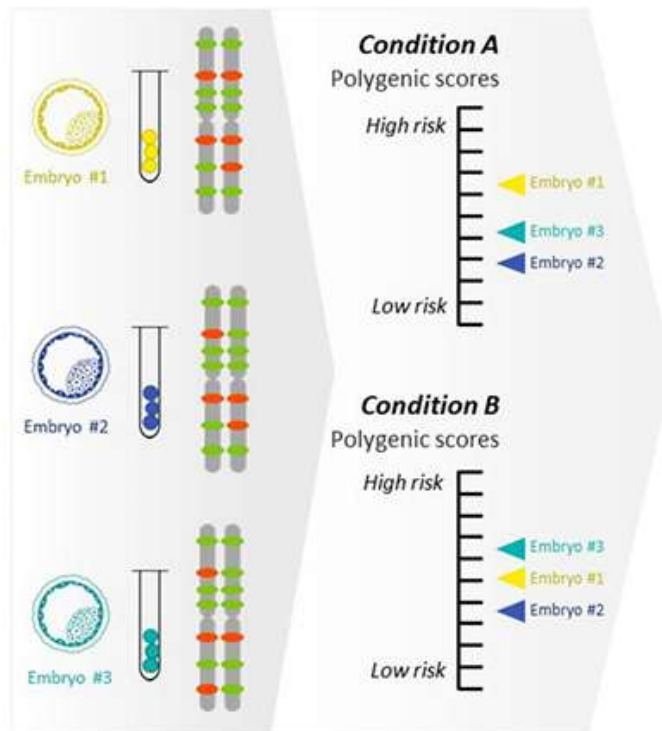
From: [Genetic association and machine learning improve the prediction of type 1 diabetes risk](#)



# Screening embryos for polygenic disease risk: a review of epidemiological, clinical, and ethical considerations

Antonio Capalbo <sup>1,2,\*</sup>, Guido de Wert<sup>3</sup>, Heidi Mertes <sup>4,5</sup>, Liraz Klausner<sup>6</sup>, Edith Coonen<sup>7,8</sup>, Francesca Spinella <sup>9</sup>, Hilde Van de Velde<sup>10,11</sup>, Stephane Viville <sup>12,13</sup>, Karen Sermon <sup>10</sup>, Nathalie Vermeulen <sup>14</sup>, Todd Lencz <sup>15,16,\*</sup>, and Shai Carmi <sup>6,\*</sup>

## Polygenic embryo screening



The principles, estimated benefits, personal and societal harms, and clinical considerations of prioritizing IVF embryos based on their risk for late-onset, polygenic conditions.

Part 3:

PRS in systems genetics

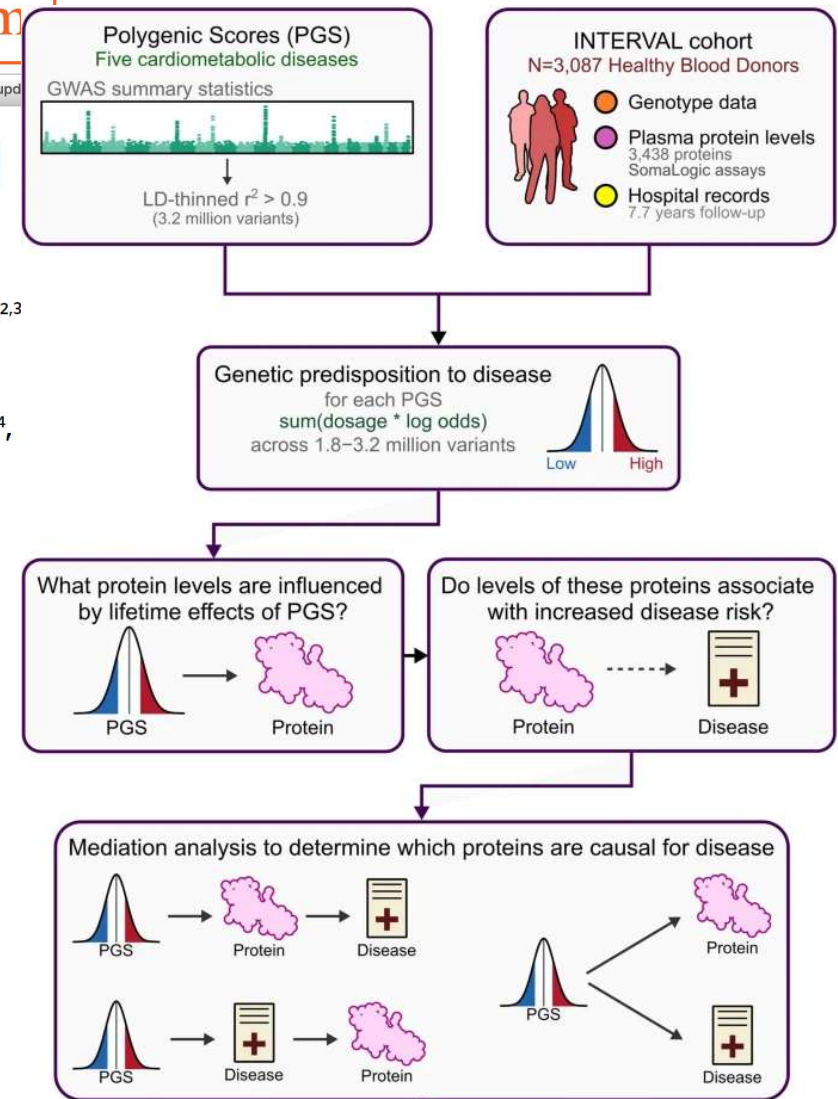




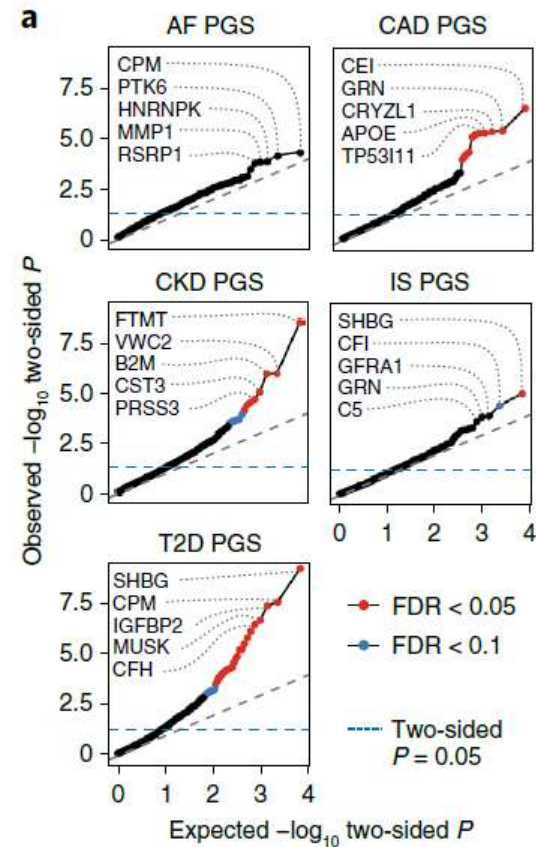
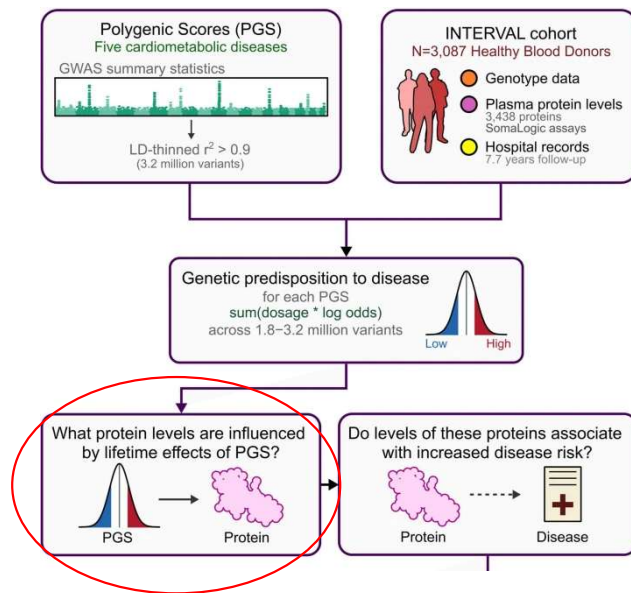
# Integrative analysis of the plasma proteome and polygenic risk of cardiometabolic diseases

Scott C. Ritchie<sup>1,2,3,4</sup> , Samuel A. Lambert<sup>1,2,3,5</sup> , Matthew Arnold<sup>1,3</sup> , Shu Mei Teo<sup>1,2</sup> , Sol Lim<sup>1,2,3</sup> , Petar Scepanovic<sup>1,2,3</sup> , Jonathan Marten<sup>1,3</sup> , Sohail Zahid<sup>6,7</sup> , Mark Chaffin<sup>1,6</sup> , Yingying Liu<sup>8,9</sup> , Gad Abraham<sup>1,2,10</sup> , Willem H. Ouwehand<sup>1,4,11,12,13,14</sup> , David J. Roberts<sup>12,14,15</sup> , Nicholas A. Watkins<sup>12</sup> , Brian G. Drew<sup>1,2,9,16</sup> , Anna C. Calkin<sup>1,2,8,16</sup> , Emanuele Di Angelantonio<sup>3,4,5,14,17</sup> , Nicole Soranzo<sup>1,4,13,14</sup> , Stephen Burgess<sup>1,3,18</sup> , Michael Chapman<sup>1,3,5,13</sup> , Sekar Kathiresan<sup>1,19</sup> , Amit V. Khera<sup>6,7,20,21</sup> , John Danesh<sup>3,4,5,13,14</sup> , Adam S. Butterworth<sup>1,3,4,5,14</sup> and Michael Inouye<sup>1,2,3,4,5,10,22</sup>

- This study demonstrated how PGS can be used to identify proteins with causal roles in disease aetiology
- To quantify each participant's relative polygenic risk of atrial fibrillation (AF), CAD, chronic kidney disease (CKD), ischaemic stroke (IS) and T2D.

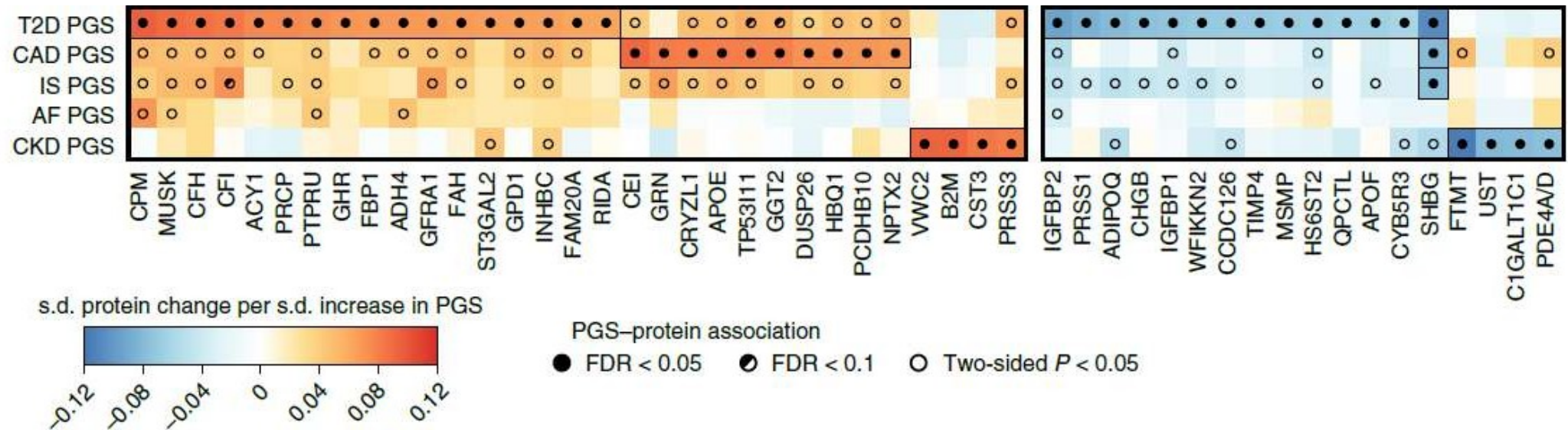


# Associations between PGS and protein levels



Quantile–quantile plots of two-sided  $P$  values from linear regression testing associations between PGS and protein levels in  $n = 3,087$  INTERVAL participants across all 3,438 tested proteins. Each plot compares the distribution of observed two-sided  $P$  values (y axes) to the distribution of expected two-sided  $P$  values under the null hypothesis for 3,438 tests (x axes) on a  $-\log_{10}$  scale. Associations were fitted using linear regression adjusting for age, sex, ten genotype principal components, sample measurement batch and time between blood draw and sample processing. The top five proteins by  $P$  value are labelled

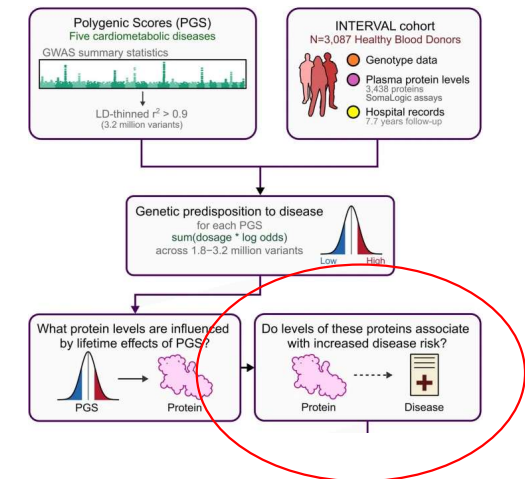
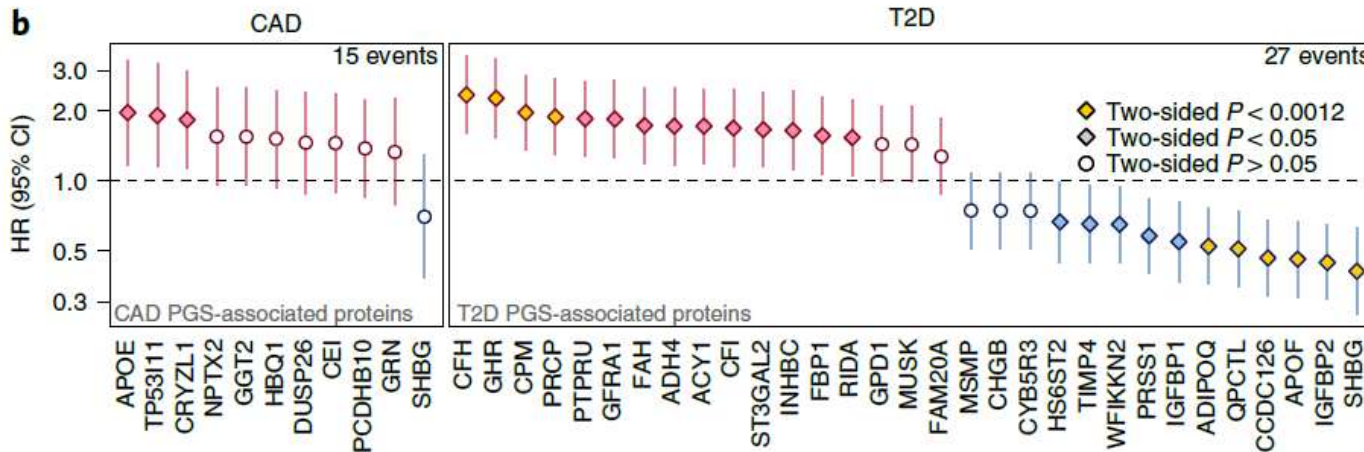
# The 49 proteins whose levels were significantly associated with at least one PGS



Each heatmap cell shows the s.d. change in protein levels per s.d. increase in PGS

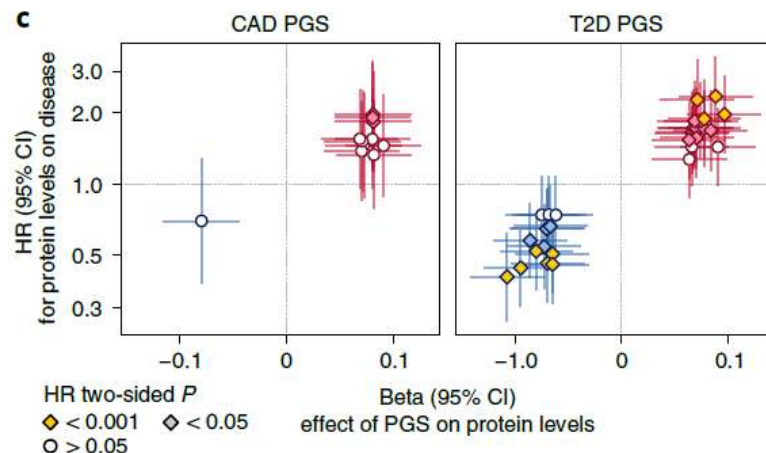


# Relations of the identified proteins to disease



The data shown correspond to the HR for CAD or T2D conferred **per s.d. increase in protein levels** (points) and its 95% CI (vertical bars).

P-values were generated using Cox proportional hazards models adjusting for age, sex, sample measurement batch and time between blood draw and sample processing.



A directional consistency between the effects of PGS on protein levels and hazard ratios (HRs) for protein levels on incident disease risk

# Causality/mediation analysis

